MILITARY SPECIFICATION SHEET

ADAPTERS, COAXIAL TO WAVEGUIDE
(SERIES N TO WAVEGUIDE - CLASS 1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the adapters described herein shall consist of this document and the latest issue of Specification MIL-A-22641

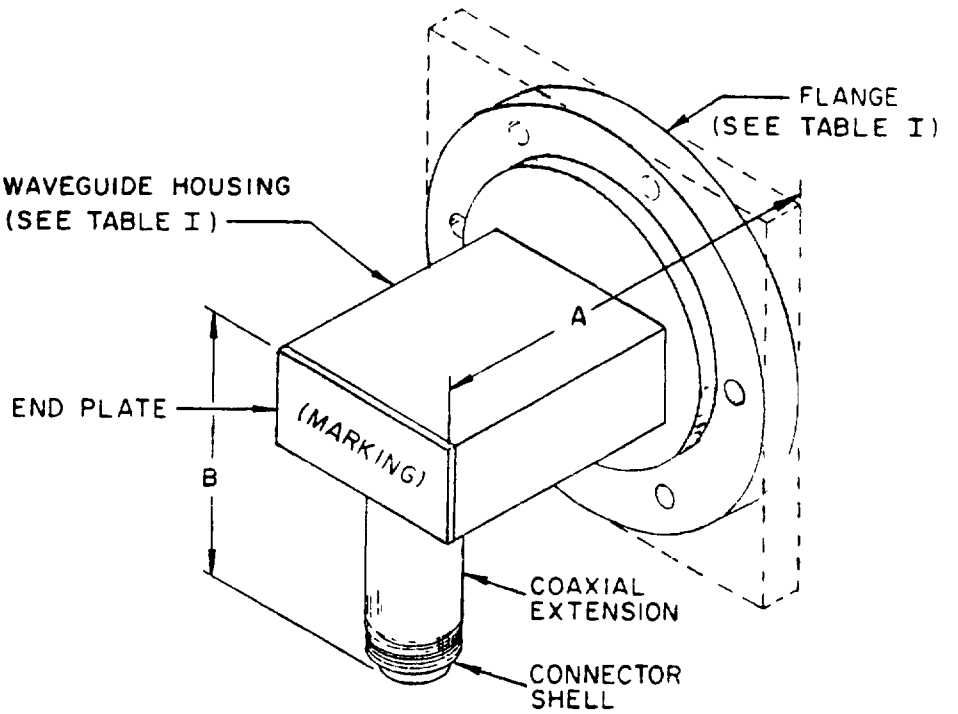


FIGURE 1 Configuration for part number M22641/16-01 thru -22.

(F) denotes changes

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.

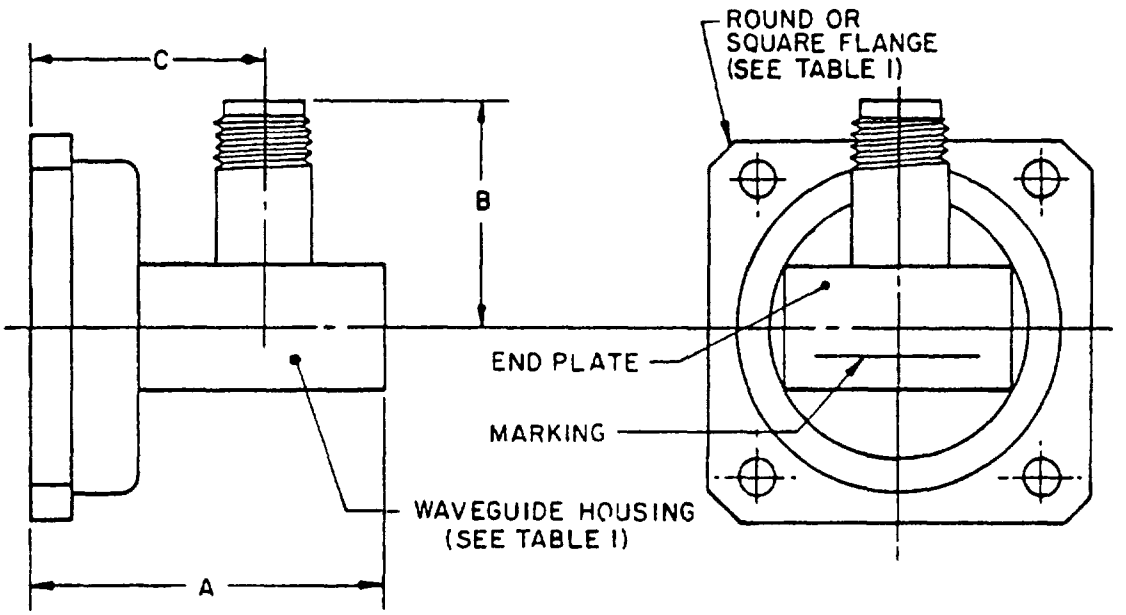


FIGURE 2 Configuration for part number M22641/16-23 thru -30

REQUIREMENTS

Design, construction, and physical dimensions See figures 1 and 2 and table I

Frequency range: See table I

VSWR: See table I.

RF insertion loss. See table I.

Radio frequency connector. The coaxial connector shall be compatible and mate with series N plug per MIL-C-39012/1.

Flange, waveguide: See table I

Housing, waveguide: See table I

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.

TABLE I Adapters, coaxial to waveguide (class I - series N to waveguide)

Dash number	Figure	Frequency range (GHz)	VSWR over entire freq range 5/ 6/	VSWR over 50% freq range 5/ 6/	RF insertion loss (dB) max 6/	Waveguide housing 7/	Flange 3/	Dimensions 1/ 2/			Waveguide and flange material
								A (max)	B (max)	C (max)	
01	1	7.05-10.0	1.25:1	1.20:1	.10	M85/1-069 M85/1-073 RG-51/U	M3922/53-002 UG-51/U Cover 3/	2.197 (55.80)	2.250 (57.15)	-	Copper alloy
02	1	5.85-8.2	1.25:1	1.20:1	.10	M85/1-063 M85/1-067 RG-50/U	M3922/55-001 UG-344/U Cover 3/	2.500 (63.50)	2.375 (60.33)	-	Copper alloy
03	1	3.95-5.85	1.25:1	1.20:1	.10	M85/1-051 M85/1-055 RG-49/U	M3922/57-002 UG-149A/U Cover 3/	3.687 (93.65)	2.437 (61.90)	-	Copper alloy
04	1	2.6-3.95	1.25:1	1.20:1	.10	M85/1-039 M85/1-043 RG-48/U	M3922/56-001 UG-53/U Cover 3/	4.062 (103.17)	4.250 (107.95)	-	Copper alloy
05	1	8.2 12.4	1.25:1	1.20:1	.20	M85/1-079 M85/1-075 RG-52/U	M3922/53-001 UG-39/U Cover 3/	2.000 (50.80)	2.218 (56.34)	-	Copper alloy
06	1	1.7-2.6	1.25:1	1.20:1	.05	M85/1-027 M85/1-031 RG-101/U	M3922/58-009 UG-435B/U Contact 3/	4.125 (104.78)	5.000 (127.00)	-	Copper alloy
07	1	1.12-1.7	1.25:1	1.20:1	.05	M85/1-015 M85/1-017 RG-69/U	M3922/58-007 UG-417B/U Contact 3/	5.250 (133.35)	7.250 (184.15)	-	Copper alloy
08	1	4.9-7.05	1.25:1	1.20:1	.10	M85/1-057 M85/1-061 RG-343/U	M3922/52-037 UG-1354/U Contact 3/	2.000 (50.80)	2.000 (50.80)	-	Copper alloy
09	1	7.05-10.0	1.25:1	1.20:1	.10	M85/1-071 RG-68/U	M3922/53-004 UG-138/U Cover 3/	2.197 (55.80)	2.250 (57.15)	-	Aluminum alloy

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.

TABLE I Adapters, coaxial to waveguide (class I - series N to waveguide). - Continued

Dash number	Figure	Frequency range (GHz)	VSWR over entire freq range 5/ 6/	VSWR over 50% freq range 5/ 6/	RF insertion loss (dB) max 6/	Waveguide housing 7/	Flange 3/	Dimensions 1/ 2/			Waveguide and flange material
								A (max)	B (max)	C (max)	
10	1	5 85-8 2	1 25 1	1.20·1	10	M85/1-065 RG-106/U	M3922/55-002 UG-441/U Cover 3/	2 500 (63 50)	2 375 (60.33)	-	Aluminum alloy
11	1	3 95-5.85	1 25 1	1 20 1	10	M85/1-053 RG-95/U	M3922/57-001 UG-407/U Cover 3/	3 687 (93 65)	2 437 (61 90)	-	Aluminum alloy
12	1	2 6-3 95	1.25·1	1 20 1	10	M85/1-041 RG-75/U	M3922/56-002 UG-584/U Cover 3/	4 062 (103 17)	4 250 (107 95)	-	Aluminum alloy
13	1	8 2-12.4	1 25 1	1 20 1	20	M85/1-077 RG-67/U	M3922/53-003 UG-135/U Cover 3/	2 000 (50 80)	2 218 (56 34)	-	Aluminum alloy
14	1	1 7-2 6	1.25 1	1 20·1	10	M85/1-029 RG-105/U	M3922/58-010 UG-437B/U Contact 3/	4 125 (104 78)	5 000 (127 00)	-	Aluminum alloy
15	1	1 12-1 7	1.25·1	1 20 1	05	M85/1-018 RG-103/U	M3922/58-008 UG-418B/U Contact 3/	5 250 (133 35)	7 250 (184 15)	-	Aluminum alloy
16	1	4 9-7 05	1 25 1	1 20 1	10	M85/1-059 RG-344/U	M3922/52-038 UG-1355/U Contact 3/	2 000 (50 80)	2.000 (50.80)	-	Aluminum alloy
17	1	7 0-11 0	1.25 1	1 20 1	20	M85/1-155 RG-320/U	M3922/70-001 UG-1493/U Cover 3/	1 515 (38 48)	1.755 (44 58)	-	Copper alloy
18	1	4 9-7,05	1.25 1	1 20 1	10	M85/1-059 M85/1-060 RG-344/U	M3922/52-038 UG-1355/U Contact 3/	3 500 (88.90)	2.188 (55 58)	-	Aluminum alloy

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.

TABLE I Adapters, coaxial to waveguide (class I - series N to waveguide) - Continued

Dash number	Figure	Frequency range (GHz)	VSWR over entire freq range 5/ 6/	VSWR over 50% freq range 5/ 6/	RF insertion loss (dB) max 6/	Waveguide housing 7/	Flange 3/	Dimensions 1/ 2/			Waveguide and flange material
								A (max)	B (max)	C (max)	
19	1	4 9- 7 05	1 25 1	1 20 1	10	M85/1-061 RG-343/U	M3922/52-037 UG-1354/U Contact 3/	3 500 (88 90)	2 188 (55 58)	-	Copper alloy
(F) 20	1	2 25- 5 5	1.30 1	1 20 1	10	M85/4-001 RG-109/U	M3922/75-01 UG-509/U Cover	6 000 (152 40)	4.000 (101 60)	-	Copper alloy
(F) 21	1	5 0- 10 0	1 40 1	1 30 1	10	M85/4-003 RG-110/U	M3922/75-03 UG-511/U Cover	3 00 (76 20)	2 75 (69 86)	-	Copper alloy
22 4/	1	7 0- 11 0	1 10 1	1 10 1	20	M85/1-155 RG-320/U	M3922/70-001 UG-1493/U 3/ Cover	1 40 (35 56)	1 10 (27 94)	-	Copper alloy
23	2	3 95- 5 85	1 065 1	1 03 1	.10	M85/1-055 RG 49/U	M3922/52-013 UG-1728/U Contact 3/	2 00 (50 80)	1 96 (30 38)	1 50 (38 10)	Copper alloy
24	2	3 95- 5 85	1 065 1	1 03 1	10	M85/1-053 RG-95/U	M3922/52-014 UG-1729/U Contact 3/	2 00 (50 80)	1 96 (30.38)	1 50 (38 10)	Aluminum alloy
25	2	7 05- 10 00	1 065 1	1 03 1	10	M85/1-073 RG-51/U	M3922/52-019 UG-1734/U Contact 3/	1 44 (36 58)	1 07 (27 18)	88 (22 35)	Copper alloy
26	2	7 05- 10 00	1 065 1	1 03 1	.10	M85/1-071 RG-68/U	M3922/52-020 UG-1735/U Contact 3/	1 44 (36 58)	1 07 (27 18)	88 (22 35)	Aluminum alloy
(F) 27 8/	2	2.60 3.95	1 04 1	---	0.05	M85/4 -017	M3922/75-06	2.50 (63.5	1.29 (33.0)	1.53 (38.86)	Aluminum alloy

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.

Ⓕ TABLE I. Adapters, coaxial to waveguide (class I - series N to waveguide). -Continued.

Dash number	Figure	Frequency range (GHz)	VSWR over entire freq. range 5/6/	VSWR over 50% freq. range 5/6/	RF insertion loss (dB) max 6/	Waveguide housing 7/	Flange 3/	Dimensions 1/2/			Waveguide and flange material
								A (max)	B (max)	C (max)	
28 8/	2	2.60-5.85	1.25 1 max	---	0.05	M85/4 -007	M3922/ 75-02	2.50 (63 5)	1.47 (37.0)	1.60 (40.6)	Aluminum alloy
29 8/	2	5.85-8.20	1 04 1	---	0.05	M85/4 -008	M3922/ 75-04	1.50 (38.1)	1.17 (30.0)	1.13 (28.7)	Aluminum alloy
30 8/	2	5 85-12.4	1 25 1 max	---	0.05	M85/4 -008	M3922/ 75-04	1.50 (38.1)	1.17 (30.0)	1.13 (28.7)	Aluminum alloy

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3/ Flange in accordance with MIL-F-3922.
4/ Connector shall be female type N (corrosion-resistant steel). Shock, vibration, and salt spray requirements are not required for this part number.
5/ A larger attenuator pad may be used for VSWR test setup (see figure 1 of MIL-A-22641) when measuring VSWR values less than 1.25 1.
6/ Swept frequency measuring technique is an acceptable alternate test method for determining VSWR and RF insertion loss values.
7/ Waveguide in accordance with MIL-W-85.
8/ Reduced height waveguides and flanges.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.

Marking. See figures 1 and 2

Part number M22641/16- (dash number from table I).

TABLE II Cross-reference of AN nomenclature to part number.

AN nomenclature	Part number M22641/16-( )	AN nomenclature	Part number M22641/16-( )
UG-397/U	01	UG-1653/U	09
UG-398/U	02	UG-1654/U	10
UG-399/U	03	UG-1655/U	11
UG-400/U	04	UG-1656/U	12
UG-446A/U	05	UG-1657/U	13
UG-952/U	06	UG-1658/U	14
UG-953/U	07	UG-1659/U	15
UG-1004/U	20	UG-1660/U	16
UG-1005/U	21	UG-1661/U	08
UG-1538/U	17	UG-1679/U	19
UG-1646/U	18		

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI, EL, GL
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AT
Navy - AS, MC

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5985-0876-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE MIL-A-22641/16F

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T13:10Z
Check the source to verify that this is the current version before use.